Chapter 12. Support Vector Machines

Lecture 9 (hyperplanes, maximal margin classifier, support vector classifier, kernels, multiclass SVM, SVM vs logistic regression)

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1 Separating Hyperplanes

1.1 Motivation

For *linearly separable* classes, find the hyperplane with the *biggest separation*. Otherwise soften "separates" and enlarge the feature space.

1.2 Hyperplanes

A flat affine subspace of dimension $p-1$:

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p = 0;$$

With $f(X) = \beta_0 + \beta^\top X$ and $y_i = \pm 1$, $f = 0$ is a **separating hyperplane** if $y_i f(x_i) > 0$ for all i; classify by $\operatorname{sign} f(x)$, with confidence $|f(x)|$.

Theorem: distance to a hyperplane

The signed distance of x to $\{f = 0\}$ is $f(x)/\|\beta\|$; if $\|\beta\| = 1$, $f(x)$ itself is the signed distance.

Proof [Supplement]. β is normal to the hyperplane. For a point x_0 on it, the distance is the length of the projection of $x - x_0$ on $\beta/\|\beta\|$: $\beta^\top(x - x_0)/\|\beta\| = (f(x) - f(x_0))/\|\beta\| = f(x)/\|\beta\|$. ■

2 The Maximal Margin Classifier

The **margin** M is the minimum distance from the observations to the hyperplane; the **optimal separating hyperplane** maximizes it:

$$\max_{\beta_0,\beta} M \quad \text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1, \qquad y_i(\beta_0 + \beta^\top x_i) \geq M \quad \forall i.$$

Theorem: equivalent convex problem

The problem is equivalent to

$$\min_{\beta_0,\beta} \frac{1}{2}\|\beta\|^2 \quad \text{subject to} \quad y_i(\beta_0 + \beta^\top x_i) \geq 1 \quad \forall i,$$

with margin $M = 1/\|\beta\|$.

Proof [Supplement]. Drop the norm constraint and write the condition as $y_i(\beta_0 + \beta^\top x_i)/\|\beta\| \geq M$ (distances, by the previous theorem). The scale of (β_0, β) is arbitrary, so fix it by $\|\beta\| = 1/M$; then the constraints become $y_i(\beta_0 + \beta^\top x_i) \geq 1$ and maximizing $M = 1/\|\beta\|$ is minimizing $\|\beta\|^2/2$, a convex quadratic program. ■

Only the points on the margin, the **support vectors**, determine the solution (Exercise 12.1).

2.1 Non-separable and noisy data

Data are usually not separable (unless $n < p$), and even then one observation can move the hyperplane dramatically. The support vector classifier maximizes a **soft margin**.

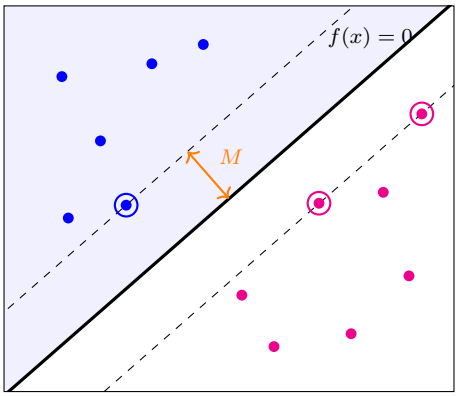


Figure 1: Maximal margin hyperplane and margins; circled: support vectors (schematic).

3 The Support Vector Classifier

Definition: soft-margin optimization

$$\max_{\beta_0,\beta,\epsilon} M \quad \text{subject to} \quad \sum_j \beta_j^2 = 1, \quad y_i(\beta_0 + \beta^\top x_i) \geq M(1-\epsilon_i), \quad \epsilon_i \geq 0, \quad \sum_{i=1}^n \epsilon_i \leq C.$$

ϵ_i are **slack variables**; $C \geq 0$ is a tuning parameter.

$\epsilon_i = 0$: correct side of the margin; $0 < \epsilon_i \leq 1$: inside the margin; $\epsilon_i > 1$: misclassified. So at most C training observations are misclassified.

C is a **budget**: larger C, wider margin; $C = 0$ is the maximal margin classifier. Chosen by CV: large C gives many support vectors, low variance, possibly high bias. Only points on or inside the margin matter, so the classifier is robust to far points, unlike LDA.

Derivation: the equivalent penalized form

Equivalently,

$$\min_{\beta_0,\beta,\xi} \frac{1}{2}\|\beta\|^2 + \gamma \sum_i \xi_i \quad \text{s.t.} \quad y_i(\beta_0 + \beta^\top x_i) \geq 1 - \xi_i, \ \xi_i \geq 0,$$

where a *large* cost γ corresponds to a *small* budget C (software usually calls γ "C" — the opposite role).

4 Support Vector Machines

4.1 Feature expansion

Enlarge the feature space, e.g. $(X_1, X_2, X_1^2, X_2^2, X_1X_2)$; a linear boundary there,

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2 + \beta_4 X_2^2 + \beta_5 X_1 X_2 = 0,$$

is a conic in the original space. High-degree polynomials get wild fast; kernels are more controlled.

4.2 Inner products and kernels

With $\langle x_i, x_{i'} \rangle = \sum_{j=1}^{p} x_{ij} x_{i'j}$, the linear support vector classifier is

$$f(x) = \beta_0 + \sum_{i=1}^{n} \alpha_i \langle x, x_i \rangle \qquad (n+1 \text{ parameters}),$$

estimated from the $\binom{n}{2}$ pairwise inner products; most $\hat{\alpha}_i$ are zero:

$$f(x) = \hat{\beta}_0 + \sum_{i \in \mathcal{S}} \hat{\alpha}_i \langle x, x_i \rangle,$$

$\mathcal{S} = \{i : \hat{\alpha}_i \neq 0\}$. A **kernel** $K(x, x_i) = \langle \Phi(x), \Phi(x_i) \rangle$ in place of $\langle x, x_i \rangle$ gives the **support vector machine**

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \hat{\alpha}_i K(x, x_i).$$

- **Polynomial kernel** of degree d: $K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^{p} x_{ij} x_{i'j}\right)^d$.
- **Radial kernel**: $K(x_i, x_{i'}) = \exp\left(-\gamma \sum_{j=1}^{p} (x_{ij} - x_{i'j})^2\right)$, $\gamma > 0$: infinite-dimensional feature space, very local; large γ, more flexible.

Example: the polynomial kernel is an inner product [Supplement]

For $p = 2$, $d = 2$: $(1 + x^\top z)^2 = 1 + 2x_1 z_1 + 2x_2 z_2 + x_1^2 z_1^2 + x_2^2 z_2^2 + 2x_1 x_2 z_1 z_2 = \langle \Phi(x), \Phi(z) \rangle$ with $\Phi(x) = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1 x_2)$.

Kernel versus enlarged feature space. Only the $\binom{n}{2}$ values $K(x_i, x_{i'})$ are needed, while degree-d monomials grow like p^d and the radial kernel has no finite feature set. K is a valid kernel iff every Gram matrix is positive semidefinite (Mercer; Exercise 12.2).

Heart data. In training, radial SVMs with large γ look nearly perfect (overfitting); on the 90 test observations they are worst, while the SV classifier, LDA and the small-γ radial SVM are similar.

5 SVMs With More Than Two Classes

- **One versus all (OVA)**: fit K two-class SVMs $\hat{f}_k(x)$, each class versus the rest; classify x to the class with the largest $\hat{f}_k(x)$.
- **One versus one (OVO)**: fit all $\binom{K}{2}$ pairwise classifiers $\hat{f}_{kl}(x)$; classify x to the class that wins the most pairwise competitions.

For moderate K, use OVO.

6 SVM versus Logistic Regression

Motivation. The SVM is hinge loss plus penalty, and the hinge loss is close to the logistic loss.

Theorem: SVM as loss plus penalty

The support vector classifier with $f(X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$ solves

$$\min_{\beta_0,\beta} \sum_{i=1}^{n} \underbrace{\max[0,\, 1 - y_i f(x_i)]}_{\text{hinge loss}} + \lambda \sum_{j=1}^{p} \beta_j^2,$$

with large λ corresponding to a large budget C.

Proof [Supplement]. In the penalized form, at the optimum each slack equals $\xi_i = \max[0, 1 - y_i f(x_i)]$ (the smallest feasible value). Substituting gives $\frac{1}{2}\|\beta\|^2 + \gamma \sum_i \max[0, 1 - y_i f(x_i)]$; divide by γ and set $\lambda = 1/(2\gamma)$. ■

The hinge loss vanishes for $y_i f \geq 1$ (hence only support vectors matter); the logistic loss $\log(1 + e^{-y_i f(x_i)})$ is never zero but small for $y_i f \gg 0$.

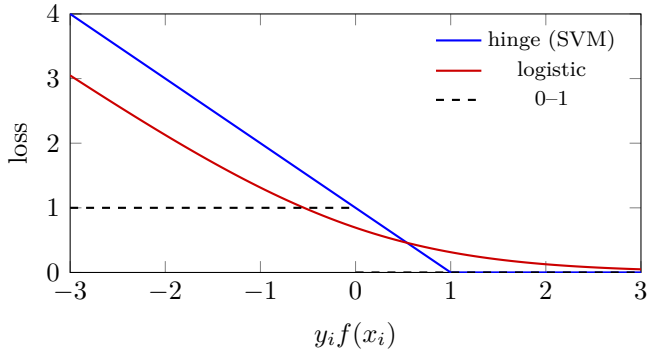


Figure 2: Hinge and logistic losses: convex surrogates of the 0–1 loss.

Which to use?

- (Nearly) separable classes: SVM (and LDA) beat logistic regression.
- Otherwise ridge-penalized logistic regression and SVM are very similar.
- For probabilities, logistic regression.
- Nonlinear boundaries: kernel SVMs (kernels also work with logistic regression and LDA, at higher cost).

Summary

- Maximal margin: $M = 1/\|\beta\|$, determined by support vectors; soft margin with budget C.
- Kernels give nonlinear SVMs through inner products; OVA/OVO for $K > 2$.
- SVM = hinge loss + ridge penalty, close to penalized logistic regression.

7 Selected exercises

Motivation. ISLR Ex. 9.3 by hand, and validity of the radial kernel.

Exercise 12.1 (A maximal margin classifier by hand; ISLR Ex. 9.3)

Seven points in $\mathbb{R}^2$: Red $(3,4),(2,2),(4,4),(1,4)$ (observations 1–4) and Blue $(2,1),(4,3),(4,1)$ (observations 5–7). Code Red as $y=+1$, Blue as $y=-1$.
(a) Find the optimal separating hyperplane and prove that it is optimal.
(b) State the classification rule, the margin and the support vectors.
(c) Show that a slight movement of observation 7 does not change the hyperplane.

Solution. We solve $\min \frac{1}{2}\|\beta\|^2$ subject to $y_i(\beta_0+\beta_1 x_{i1}+\beta_2 x_{i2})\geq 1$.

(a)

Step 1 (a lower bound). The constraints of observations 2, 5 and 6 read

$$\beta_0+2\beta_1+2\beta_2\geq 1, \qquad -\beta_0-2\beta_1-\beta_2\geq 1, \qquad -\beta_0-4\beta_1-3\beta_2\geq 1.$$

Adding the first two gives $\beta_2\geq 2$; adding the first and third gives $-2\beta_1-\beta_2\geq 2$, so $\beta_1\leq -1-\beta_2/2\leq -2$. Hence every feasible β has $\|\beta\|^2=\beta_1^2+\beta_2^2\geq 8$, with equality only for $\beta_1=-2$, $\beta_2=2$.

Step 2 (the intercept). With $\beta_1=-2$, $\beta_2=2$, the first constraint gives $\beta_0\geq 1$ and the second $\beta_0\leq 1$, so $\beta_0=1$.

Step 3 (feasibility). $f(x)=1-2x_1+2x_2$ takes the values $3,1,1,7$ at the Red points and $-1,-1,-5$ at the Blue points, so all constraints hold. By Steps 1–2 it is the unique optimum. The hyperplane is $1-2X_1+2X_2=0$, i.e. $0.5-X_1+X_2=0$.

(b) Classify to Red if $0.5-X_1+X_2>0$ and to Blue otherwise. The margin is $M=1/\|\beta\|=1/\sqrt{8}\approx 0.354$. The support vectors are the points with $y_i f(x_i)=1$: observations 2, 3, 5 and 6.

(c) Observation 7 has $y_7 f(x_7)=5$. If it moves by δ with $\|\delta\|<\sqrt{2}$, then $|f(x_7+\delta)-f(x_7)|=|\beta^\top\delta|\leq\sqrt{8}\|\delta\|<4$, so its constraint still holds and (β_0,β) remains feasible. The lower bound of Step 1 and Step 2 used only observations 2, 5 and 6, which did not move; so the same hyperplane is still the unique optimum. ■

Exercise 12.2 (The radial kernel is positive semidefinite)

Show that for any $x_1,\dots,x_n\in\mathbb{R}^p$ and $\gamma>0$ the matrix $K=\left[\exp(-\gamma\|x_i-x_j\|^2)\right]_{i,j}$ is positive semidefinite.

Solution [Supplement].

Step 1. The Gram matrix $G=[x_i^\top x_j]$ is PSD: $c^\top Gc=\|\sum_i c_i x_i\|^2\geq 0$.

Step 2 (Schur product). If A,B are PSD, so is the entrywise product $A\circ B$. Write $B=\sum_k \lambda_k v_k v_k^\top$ with $\lambda_k\geq 0$; then

$$c^\top(A\circ B)c=\sum_k\lambda_k\sum_{i,j}(c_i v_{ki})A_{ij}(c_j v_{kj})=\sum_k\lambda_k(c\circ v_k)^\top A(c\circ v_k)\geq 0.$$

Step 3. By induction, every entrywise power $G^{\circ m}$ is PSD ($G^{\circ 0}=\mathbf{1}\mathbf{1}^\top$). Hence $E_M=\sum_{m=0}^{M}\frac{(2\gamma)^m}{m!}G^{\circ m}$ is PSD, and $E_M\to E=[e^{2\gamma x_i^\top x_j}]$ entrywise; since $c^\top E_M c\geq 0$ for every M, also $c^\top Ec\geq 0$.

Step 4. $\|x_i-x_j\|^2=\|x_i\|^2-2x_i^\top x_j+\|x_j\|^2$ gives $K=DED$ with $D=\operatorname{diag}(e^{-\gamma\|x_i\|^2})$, so $c^\top Kc=(Dc)^\top E(Dc)\geq 0$.

By Mercer, the radial kernel is an inner product in some feature space. ■